

Zainab Vohra

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Education

RUTGERS UNIVERSITY

School of Arts and Sciences

Bachelors of Science, in Computer Science, Bachelor of Arts in Cognitive Science

New Brunswick, NJ

Graduation Date: Spring 2028

- GPA: 3.7
- Key Courses: Principles of Computer Science, Introduction to Computer Science, Data Structures & Algorithms, Linear Algebra, Discrete Structures I, Discrete Structures II, Calc I, Calc II, Introduction to Cognitive Science, Computer Architecture

Technical Skills

- **Programming Languages:** Python, Java, C, C++, Assembly, Javascript, CSS, HTML
- **Operating Systems:** Windows, MacOS, Linux

Experience

Kumon

Center Assistant

Hackettstown, NJ

February 2024 – January 2026

- Tutoring students in math and reading, from early learners to high school level.
- Grade assignments, track student progress, and provide feedback to improve accuracy and speed.
- Support early learners with foundational skills such as letter/number recognition and fine motor skills.
- Assist with parent communication, front-desk tasks, and day-to-day center operations.

Township of Randolph

Revenue Clerk

- Maintained and updated financial records to ensure accurate account balances and payment histories
- Utilized government software/payment processing systems to efficiently manage records and transactions
- Provided residents service by answering questions regarding tax and utility accounts and payment status

Projects

Linked List Train Car Project (Java)

December 2024

- Designed and implemented a custom doubly linked list data structure to simulate train cars, supporting dynamic insertion, deletion, and traversal operations.
- Applied principles of data structures, modular design, and debugging with test-driven development (JUnit + Hamcrest).

Matryoshka & Quadratic Koch Fractal Project (Java)

July 2025

- Implemented recursive algorithms to generate fractal patterns, including Matryoshka nesting and the Quadratic Koch curve.
- Utilized the StdDraw library for custom graphics, enabling visualization of recursion depth, symmetry, and geometric transformations.
- Reinforced problem-solving skills in recursion, object-oriented design, and algorithmic thinking through interactive visual outputs.

BMR & Calorie Tracking Program (Python)

June 2024

- Built a console-based application to calculate Basal Metabolic Rate (BMR), daily caloric intake, and projected weight change using user-provided inputs.
- Applied control structures, loops, and mathematical formulas to provide personalized health insights over multiple days.

Leadership and Activities

Islamic Society of New Jersey

Flanders, NJ

Volunteer Teacher & Program Assistant

September 2022 – June 2024

- Taught and mentored children in a classroom setting, supporting their learning and engagement.
- Communicated with parents to provide updates and ensure a welcoming, organized environment.
- Coordinated lunch and recess activities, helping maintain safety, smooth transitions, and positive interactions among students.

Rutgers University

September 2024

- Women in Computer Science member (WiCS)
- Muslim Student Association member (MSA)